MATH 125: COMBINATORIAL ALGEBRAIC GEOMETRY

SAIR SHAIKH

1. Intro to Toric Geometry:

There are 2 mottos in toric geometry.

(1) Toric geometry is the study of solutions to polynomoial equations in the presence of a large torus.
(2) The action of a torus should give combinatorial/algebraic structure.

To make sense of this, we need to understand what solutions to polynomial equations are and what is a large torus.

A geometric torus is $\underbrace{S^1 \times S^1 \times \cdots \times S^1}_{n \text{ times}}$. An algebraic torus is $(\mathbb{C}^\times)^\times$ or, more generally, $(k^\times)^n$ for some field k.

We will now do an interlude in basic algebraic geometry.

Let k be a field. We call $\mathbb{A}^n_k = k^n$ the affine n-space. Next, we define algebraic sets and the Zariski topology.

We give some examples of algebraic sets: The vanishings of $x^2 + y^2$, xy, $x^2y - y$, and $z^2 - x^2 - y^2$ in the appropriate affine spaces.

Definition 1.1. *An affine algebraic vareirt is an algebraic subset of $\mathbb{A}^n_k$ such that it is irreducible in the Zariski topology.*

Theorem 1.2 (Nullstellensatz)**.** *There is an order reversing bijection:*

$$\{\textit{Algebraic varieties in } \mathbb{A}^n_k\} \leftrightarrow \{I \subseteq k[x_1, \cdots, x_n] \textit{ such that } I \textit{ is prime. }\}$$

Coming back to toruses, note that $k^\times \subset \mathbb{A}^1$ is a torus living in our affine space. Thus, A^1 is a toric variety. Similarly, $(k^\times)^n = \mathbb{A}^n \setminus \mathbb{V}(x_1 \cdots x_n) \subseteq \mathbb{A}^n$ is an big torus. Hence, $\mathbb{A}^n$ is a toric variety.

Definition 1.3. *If $X \subseteq \mathbb{A}^n$ is an algebraic set, the coordinate ring is:*

$$k[X] = k[x_1, \cdots, x_n]/I_X$$

where $I_X = \langle f \in k[x_1, \cdots, x_n] | f(p) = 0 \, \forall x \in X \rangle$.

Lemma 1.4. *If $f \in S = k[x_1, \cdots, x_n]$, then the set $D(f) = \mathbb{A}^n \setminus \mathbb{V}(f)$ is such that $k[D(f)] \equiv S\left[\frac{1}{f}\right]$*

Then, as examples, we get that the large tori:

(1) $k[(k^\times)] = k[x, x^{-1}]$

(2) $k[(k^\times)^n] = k[x_1, x_1^{-1}, \cdots, x_n, x_n^{-1}]$

Now we can define what a toric variety is.

Definition 1.5. *An affine toric vareity is an affine algebraic variety* X *together with an open dense torus* $T \subseteq X$ *such that the action of* T *on itself extends to* X*. That is, there is an action* $k^\times \times \mathbb{A}^1 \to \mathbb{A}$ *that restricts to the torus' action on itself on* $k^\times \times k^\times$*.*

We now discuss what the action of the torus means. Let $T = (k^\times)$. Then T is also a group with multiplication $(s, t) \mapsto st$ and inverse $t \mapsto t^{-1}$.

Example 1. *Let* $k^\times \subset \mathbb{A}^1$*. Then,* $k^\times$ *acts on itself. We extend this by letting the action on* 0 *be such that* $t \times 0 = 0$ *for all* $t \in k^\times$*.*

Now we make all of this precise. We have the following definition:

Definition 1.6. *An algebraic group is an algebraic variety* G *together with maps:*

$$\times : G \times G \to G \qquad (-)^{-1} : G \to G$$

that are regular and satisfy the usual group axioms.

Example 2. $G = (k^\times)^n$ *with coordinate with multiplication and inverses.*

Definition 1.7. *A split torus over a field* k *is algebraic group* G *isomorphisc over* k *to* $(k^\times)^n$ *for some* n*.*

Definition 1.8. *An algebraic torus over a field* k *is an algebraic group* T *such that after an extension,* $T \times_k F \cong (F^\times)^n$*.*

Remark 1.9. *To motivate this definition, consider* S^1*. We want* S^1 *to be a torus. That is,* $S^1 = \mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$ *is not* $(\mathbb{R}^\times)^n$ *for any* n*. However, if we consider* $S^1 = \mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}$*, we can factor the polynomial to write it as* $\mathbb{V}(zw - 1)$*. Considering* $\mathbb{C}[z, w]/\langle zw - 1\rangle \cong \mathbb{C}[x, x^{-1}]$*.*

Definition 1.10. *An affine toric variety is an affine algebraic variety* X *together with a torus* $T \subseteq X$ *that is open and dense together with an algebraic action:*

$$\begin{tikzcd} T \times X \arrow[r] & T \\ T \times T \arrow[r] \arrow[u] & T \arrow[u] \end{tikzcd}$$

Definition 1.11. *Let* $X \subseteq \mathbb{A}^n$ *and* $Y \subseteq \mathbb{A}^m$ *be affine algebraic varieties. A morphism is globally given by polynomials* $(f_1, \cdots, f_m)$ *that restrict to a map from* X *to* Y*.*

Lemma 1.12. $\Phi : X \to Y$ *is a morphism of algebraic varities if and only if there is an induced (pullback) map of rings:*

$$\Phi^* : k[Y] \to k[X]$$

Example 3 (Toric Varieties)**.** .

(1) $X = (k^\times)^n$
(2) $X = \mathbb{A}^1$
(3) $X = \mathbb{A}^n$*. Extend entry-wise.*
(4) $X = \mathbb{V}(x^3 - y^2)$

2. Intro to Algebraic Geometry:

We are taking a detour and doing classical AG fast but properly before going into Toric Geometry.

Let k be a field. You can imagine $k = \mathbb{C}$ for most of this.

Definition 2.1. *An algebraic set is a subset:*

$$\mathbb{V}(f_1, \cdots, f_t) = \{x \in \mathbb{A}^n_k | f_1(x) = f_2(x) = \cdots = f_t(x) = 0\}$$

where $f_i \in k[x_1, \cdots, x_n]$.

Given an ideal $I \subseteq k[x_1, \cdots, x_n]$. We define,

$$\mathbb{V}(I) = \{x \in \mathbb{A}^n_k | f(x) = 0 \, \forall f \in I\}$$

Remark 2.2. *By the Hilbert Basis theorem,*

$$I = \langle f_1, \cdots, f_t \rangle$$

because the ring is noetherian. Thus, $\mathbb{V}(I) = \mathbb{V}(f_1, \cdots, f_t)$ *is an algebraic set.*

Definition 2.3. *If* $V \subseteq \mathbb{A}^n$ *is an algebraic set. We define,*

$$\mathbb{I}(V) = \{f \in k[x_1, \cdots, x_n] \, | f(x) = 0 \, \forall x \in V\}$$

Lemma 2.4. $\mathbb{I}(V)$ *is an ideal*

Lemma 2.5. *If* $V \subseteq \mathbb{A}^n_k$*, then*

$$\mathbb{V}(\mathbb{I}(V)) = V$$

Proof. **TODO:** Do this. □

Warning:

$$\mathbb{I}(\mathbb{V}(I)) \neq I$$

Example 4. *(1)* $k = \mathbb{R}$*. Let* $I = \langle x^2 + 1 \rangle$*. Then* $\mathbb{V}(I) = \emptyset$ *and* $\mathbb{I}(\emptyset) = \mathbb{R}[x]$*.*
(2) $k = \overline{k}$ *and* $I = \langle x^2 \rangle \subseteq \mathbb{A}^1$*.* $\mathbb{V}(I) = \{0\}$ *and* $\mathbb{I}(\{0\}) = \langle x \rangle$*.*

Proposition 2.6. *If* $k = \overline{k}$ *and* $I \subseteq k[x_1, \cdots, x_n]$ *is an ideal*

$$\mathbb{I}(\mathbb{V}(I)) = \{f \in k[x_1, \cdots, x_n] | \exists m : f^m \in I\} = \sqrt{I}$$

Theorem 2.7 (Hilbert's Nullstellensatz)**.** *Assume* $k = \overline{k}$*. Then,*

(1) Let $m \subseteq k[x_1, \cdots, x_n]$ *be an ideal.* m *is maximal if and only if* $m = \langle x_1 - a_1, \cdots, x_n - a_n \rangle$ *for some* $(a_1, \cdots, a_n) \in \mathbb{A}^n_k$*.*
(2) There is an inclusion reversing bijection between algebraic subsets of $\mathbb{A}^n_k$ *and radical ideals in* $k[x_1, \cdots, x_n]$ *given by* $\mathbb{I}$ *and* $\mathbb{V}$*.*

Definition 2.8. *A polynomial function on* $\mathbb{A}^n_k$ *is a function of the form* $f : \mathbb{A}^n_k \to k$ *given by* $\overline{x} \mapsto f(\overline{x})$ *where* $f \in k[x_1, \cdots, x_n]$*.*

Now we wanna define polynomial functions on algebraic sets. We do this as follows:

Definition 2.9. *Let $V \subseteq \mathbb{A}^n$ is an algebraic set. A function $f : V \to k$ is a polynomial function if and only if there exists a polynomial function $F : \mathbb{A}^n \to k$ such that the following commutes,*

$$\begin{tikzcd} \mathbb{A}^n \arrow[dr, "F"] & \\ V \arrow[u] \arrow[r, "f"] & k \end{tikzcd}$$

The choice of lift is not unique.

Example 5. *Let $V = \mathbb{V}(y - x^2)$. Let $F(x, y) = y$ and $G(x, y) = 2y - x^2$. Restricted to V they are the same function as $2y - x^2 = y + (y - x^2) = y$.*

Example 6. *Consider $F(x, y) = e^{x^2+y^2}$ and $V = \mathbb{V}(x^2 + y^2 - 1)$. Then $F|_V(x, y) = e$, which is a polynomial.*

Example 7. *Consider $V = \mathbb{V}(xy - 1) \subseteq \mathbb{A}^2$. Consider $F(x, y) = \frac{1}{x}$. This is not a function on $\mathbb{A}^2$ but it is a function on V. Moreover, it is a polynomial function because $\frac{1}{x} = y$ on V.*

Definition 2.10. *The coordinate ring of an algebraic set $V \subseteq \mathbb{A}^n$, $k[V]$ is the set of polynomials functions on V with pointwise multiplication and addition.*

Lemma 2.11. *If $V \subset \mathbb{A}^n$ is an algebraic set, then the coordinate ring $k[V]$ is isomorphic to $k[x_1, \cdots, x_n]/\mathbb{I}(V)$.*

Proof. We have a map,

$$k[x_1, \cdots, x_n] \twoheadrightarrow k[V]$$

sending $f \mapsto [x \mapsto f(x)]$. This fits in the exact sequence,

$$0 \to \mathbb{I}(V) \to k[x_1, \cdots, x_n] \to k[V] \to 0$$

□

Lemma 2.12. *If $k \supseteq \mathbb{R}$, then an algebraic set is closed in the Euclidean topology $\mathbb{A}^n_k$.*

Proof. The finite intersection of closed sets is closed. □

Definition 2.13. *The Zariski topology $\mathbb{A}^n_k$ has closed sets equal to all the algebraic sets.*

Furthermore, we set the Zariski topology on an algebraic set $V \subseteq \mathbb{A}^n$ to be the subspace topology $V \cap Z$ where $Z \subseteq \mathbb{A}^n_k$ is an algebraic set.

Given a set $S \subseteq \mathbb{A}^n_k$, we have $\overline{S} = \mathbb{V}(\mathbb{I}(S))$.

Definition 2.14. *A morphism of algebraic sets $V \subset \mathbb{A}^n, W \subset \mathbb{A}^m$ is a function $f : V \to W$ such that,*

$$\begin{tikzcd} \mathbb{A}^n \arrow[r, "F"] & \mathbb{A}^m \\ V \arrow[u] \arrow[r, "f"] & W \arrow[u] \end{tikzcd}$$

where $F : (x_1, \cdots, x_n) \mapsto (F_1(\overline{x}), \cdots, F_m(\overline{x}))$ with $F_1, \cdots, F_m \in k[x_1, \cdots, x_m]$.

Definition 2.15. *Algebraic sets are isomorphisc if and only if there exists morphisms that are mutually inverses.*

Remark 2.16. *These functions are continuous in the Zariski topology because closed sets are given by the vanishing of polynomials and pulling back polynomials by polynomials still give polynomials.*

Remark 2.17. *These maps are not closed. Consider* $F(x, y) \mapsto y$. *Then* $\mathbb{V}(xy - 1) \subseteq \mathbb{A}^2$ *maps to* $\mathbb{A}^1 \setminus \{0\} \subset \mathbb{A}^1$ *not closed.*

Definition 2.18. *A finitely generated reduced* k*-algebrai is* $k[x_1, \cdots, x_n]/I$ *where* I *is radical.*

Theorem 2.19 (Nullstellensatz II)**.** *Assume* $k = \overline{k}$.

(1) If $f : V \to W$ *is a morphism of algebraic sets then there is a morphism of* k*-algebras,*

$$F^{\#} : k[V] \to k[W]$$
$$[g : W \to k] \mapsto [g \circ f : V \to k]$$

(2) If $I = \mathbb{I}(V)$ *and* $J = \mathbb{I}(W)$ *and there exists a map* $\sigma : k[x_1, \cdots, x_m]/J \to k[x_1, \cdots, x_n]/I$ *then, pick* $F_i \in k[x_1, \cdots, x_n]$ *such that* $\sigma(y_i) = F_i$ *mod* I. *Then the map,*

$$\overline{F} : \mathbb{A}^n \to \mathbb{A}^m$$
$$x \mapsto (F_1(x), \cdots, F_m(x))$$

restricts to a map $V \to W$ *and this restriction is independent of choice.*

(3) Moreover, $F^{\#} = \sigma$.

(4) $V \cong W \iff k[V] \cong k[W]$.

(5) There is an (anti)-equivalence of categories between categories of algebraic sets and their morphisms and finitely generated reduced k*-algebras.*

We have now defined the major correspondence that's studied in a first semester graduate course in classical AG. The work we have still left is translating from one language to another, i.e. dimension, singularities, etc.

We have a glaring problem because we are working in an ambient affine space. So, we expand first to projective space.

Definition 2.20. *Projective* n*-space/*k *is*

$$\mathbb{P}^n_k = \{ \text{ one-dim subspaces of } k^{n+1} \}$$

Lemma 2.21.

$$\mathbb{P}^n = (k^{n+1} \setminus \{0\})/$$

where $(a_0, \cdots, a_n) \sim (b_0, \cdots, b_n)$ *if and only if there exists* $\lambda \in k^\times$ *such that one is a scalar multiple of the other.*

Next, we define the homogeneous coordinates.

Definition 2.22. *The coordinates on* $\mathbb{P}^n$ *can be given as* $[a_0 : a_1 : \cdots : a_n]$. *This is only unique up to scaling by* $k^\times$. *In other words,* $(a_0, \cdots, a_n)$ *and* $(b_0, \cdots, b_n) \in \mathbb{A}^{n+1}$ *define the same point if the* $2x2$ *minors of*

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \end{pmatrix}$$

are all 0.

Example 8. *Consider* $\mathbb{P}^1_{\mathbb{R}}$. *We drew the picture where we picked the reference line* $x_0 = 1$ *and mapped all lines to the point on this line, i.e. a line goes to* $(1, \lambda)$ *where* λ *is its slope. We got a map,*

$$\mathbb{P}^1_{\mathbb{R}} \to \mathbb{R} \cup \{\infty\}[x_0 : x_1] \mapsto \begin{cases} \frac{x_1}{x_2} & x_0 \neq 0 \\ \infty & x_0 = 0 \end{cases}$$

Thus, we got $\mathbb{P}^1$ *as a circle. Then, we did the same thing for* $\mathbb{P}^2$ *showing* $\mathbb{P}^2 \cong \mathbb{R}^2 \cup \mathbb{P}^1$.

Lemma 2.23. *For* $i = 0, \cdots, n$,

$$U_i = \{[x_0 : \cdots : x_n] \subseteq \mathbb{P}^n_k | x_i = 0\}$$

(1) $\psi_i : \mathbb{P}^n_k \to \mathbb{A}^n \cup \mathbb{P}^{n-1}$ *is a bijection (details)*
(2) $\psi_i|_{U_i}$ *is a bijection.*
(3) $\mathbb{P}^n = \bigcup_i U_i$
(4) $\psi_j \circ \psi_i^{-1} : \psi_i(U_i \cap U_j) \to \psi_j(U_i \cap U_j)$ *is a rational function in the coordinates on* $\mathbb{A}^n$.

Example 9. *Explicltly showed that the functions compose to a rational function.*

Theorem 2.24 (Fact.)**.** *If* $k = \mathbb{R}, \mathbb{C}$ *then* $\mathbb{P}^n_{\mathbb{R}}$ *or* $\mathbb{P}^n_{\mathbb{C}}$ *are compact* $\mathbb{R}$ *or* $\mathbb{C}$ *manifolds.*

We cannot define the algebraic sets for $\mathbb{P}^n$ using global functions. For instance, $\mathbb{P}^n_{\mathbb{C}} \to \mathbb{C}$ every holomorphic function is constant.

Definition 2.25. *A polynomial* $f \in k[x_0, \cdots, x_n]$ *is homogenenous* $\iff$ *every term of* f *has the same degree* $\iff$ $f(\lambda x) = \lambda^{\deg(f)} f(x)$ *for* $\lambda \in k^\times$.

Now, we are defining functions.

Example 10. $\mathbb{P}^1_{\mathbb{C}} \to \mathbb{C}$ *given by* $[x : y] \mapsto x$ *is homogenous but is not well-defined.*

Definition 2.26. *A projective algberaic set is a subset* $V \subset \mathbb{P}^n_k$ *of the form,*

$$V = \mathbb{V}(f_1, \cdots, f_t) = \{x \in \mathbb{P}^n : f_1(x) = \cdots = f_t(x) = 0\}$$

for $f_i \in k[x_0, \cdots, x_n]$ *homogeneous. If* $I \subset k[x_0, \cdots, x_n]$ *is a homogeneous ideal then*

$$\mathbb{V}(I) = \{x \in \mathbb{P}^n : f(x) = 0 \, \forall f \in I\}$$

Definition 2.27. *If* $V \subseteq \mathbb{P}^n$ *is a projective algebraic set,*

$$\mathbb{I}(V) = \langle f \in k[x_1, \cdots, x_n] | \, f \text{ hom. and } f(x) = 0 \, \forall x \in V \rangle$$

Theorem 2.28 (Proj. Nullstellensatz)**.** *Assume* $k = \overline{k}$.

(1) If $I \subseteq k[x_0, \cdots, x_n]$ *is a homogeneous ideal,*

$$\mathbb{V}(I) = \emptyset \iff \langle x_0, \cdots, x_n \rangle^m \subseteq I)$$

for some m.
(2) $\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$.
(3) There is a bijection between projective algebraic subsets in $\mathbb{P}^n$ *and homogeneous radical ideals* $k[x_0, \cdots, x_n]$ *not equal to* $\langle x_0, \cdots, x_n \rangle$.

Beginning day 3 (Apr 3).

Definition 2.29. *The homogenous coordinate ring of a projective algebraic set* $V \subseteq \mathbb{P}^n_k$ *is*

$$k[V] = k[x_0, \cdots, x_n]/\mathbb{I}(V)$$

Warning: These are not functions on V.

We define the Zariski topology the same way.

Definition 2.30. *The Zariski topology on* $\mathbb{P}^n_k$ *has projective algebraic sets as its closed sets. The topology on an proj. alg. set is the subspace topology.*

Definition 2.31. *A morphism* $F : V \to W$ *of projective algebraic sets* $V \subseteq \mathbb{P}^n_k$ *is a function such that* $\forall p \in V$ *there exists an open set* $p \in U \subseteq V$ *and polynomials* $F_1, \cdots, F_m \in \mathbb{C}[x_0, \cdots, x_n]$ *homogeneous of degree* d *such that:*

$$F|_U : U \to \mathbb{P}^m$$

An isomorphism is a function with an inverse function.

Example 11. *Let* $C = \mathbb{V}(zx - y^2)$ *There is map* $F : C \to \mathbb{P}^1$ *defined by,*

$$[x : y : z] \mapsto \begin{cases} [x : y] & x \neq 0 \\ [y : z] & z \neq 0 \end{cases}$$

If $x \neq 0$ *and* $z \neq 0$*, then* $y \neq 0$*. Then,*

$$[x : y] = [xy : y^2] = [xy : xz] = [y : z]$$

Next, take the map,

$$\mathbb{P}^1 \to \mathbb{C} \qquad [s : t] \mapsto [s^2 : st : t^2]$$

This is well-defined. It is a morphism (given as polynomial globally). We can check these are inverses.

$$[s : t] \mapsto [s^2 : st : t^2] \mapsto \begin{cases} [s^2 : st] = [s : t] & s \neq 0 \\ [st : t^2] = [s : t] & t \neq 0 \end{cases}$$

Thus, these are isomorphisms.

However, note that:

$$k[\mathbb{P}^1] = k[s, t] \ncong k[C] = k[x, y, z]/\langle xz - y^2 \rangle$$

The projective coordinate ring depends on the embedding!

Recall, a locally closed set in a topological space X is a closed subset of an open set. That is, it is given as $U \cap Z$ for U open and Z closed.

Definition 2.32. *A quasiprojective variety is a locally closed subset of* $\mathbb{P}^n_k$*.*

Example 12. *The following are quasiprojective varieties.*

(1) Open subsets of $\mathbb{P}^n_k$*.*
(2) Projective algberaic sets.
(3) Algebraic sets. (Some things need to be shown, we show them next.)

Take $V \subseteq \mathbb{A}^n_k$ and $V = \mathbb{V}(f_1, \cdots, f_t)$. Define,

$$f_i^h = x_0^{\deg(f_i)} f_i\left(\frac{x_1}{x_0}, \cdots, \frac{x_n}{x_0}\right)$$

Then $f^h(x_0, \cdots, x_n) = 0 \iff f_i(x_1, \cdots, x_n) = 0$.

Example 13. $(k^\times)^n$ *is quasiprojective. It is* $\mathbb{A}^n \setminus \mathbb{V}(x_1, \cdots, x_n)$.

Definition 2.33. *A function* $\Phi : V \subset \mathbb{P}^n \to W \subset \mathbb{P}^m$ *is a morphism of quasiprojective varieties if and only if* $\forall p \in V$ *there exists* $F_1, \cdots, F_m$ *homg. of degree* d *in* $k[x_0, \cdots, x_n]$ *such that,*

$$\Phi|_U : x \mapsto [F_0(x), \cdots, F_m(x)]$$

Example 14. *Take the sets* $V = \mathbb{V}(xy - 1)$ *(closed in affine space) and* $W = \mathbb{A}^1 \setminus \{0\}$ *(open in affine space). First, we can homogenize to consider these in projective space. That is these live in* $\mathbb{P}^2$ *and* $\mathbb{P}^1$ *and are quasiprojective varieties.*

$$\begin{tikzcd}
\mathbb{P}^2 & \mathbb{P}^1 \arrow[l] \\
\{z \neq 0\} \cap \mathbb{V}(xy - z^2) \arrow[u] & \mathbb{A}^1 \setminus \{0\} \arrow[l] \arrow[u]
\end{tikzcd}$$

TODO: *Complete this.*

Definition 2.34. *A quasiprojective variety* $V \subseteq \mathbb{P}^m_k$ *is affine if and only if there's some algebraic set* $W \subseteq \mathbb{A}^n_k$ *such that* $V \cong W$ *as quasiprojective varieties.*

If V is a quasiprojective variety that is affine

$$k[V] := k[W]$$

for W.

If $V \subseteq \mathbb{A}^n_k$ and $W \subseteq \mathbb{A}^n_k$. We need to show,

$$V \cong_{qp} W \iff V \cong_{AS} W$$

Lemma 2.35. *Let* $V = \mathbb{V}(f_1, \cdots, f_t) \subseteq \mathbb{A}^n$ *be an algebraic set and* $g \in k[V]$ *be some function. Let* $V \setminus \mathbb{V}(g) =: U_f$ *is affine. Moreover,*

$$k[U_f] \equiv k[V][g^{-1}]$$

Proof.

$$\mathbb{A}^n \to \mathbb{A}^{n+1}$$

$$(x_1, \cdots, x_n) \in U_f \mapsto \left(x_1, \cdots, x_n, \frac{1}{g(x_1, \cdots, x_n)}\right)$$

Make sure this lands in the right place. □

Definition 2.36. *A quasiprojective variety* $V \subseteq \mathbb{P}^m_k$ *is projective if and only if there's some projective algebraic set* $W \subseteq \mathbb{P}^n_k$ *such that* $V \cong W$ *as quasiprojective varieties.*

Proposition 2.37. *Every quasiprojective variety is covered by affine varieties.*

(April 7th)

Definition 2.38. *Let $U \subseteq V$ be open inside an affine variety and $U \to k$ some function. Then,*

(1) For $p \in U$, f is regular at p if and only if $\exists p \in U' \subset U$ open and $g, h \in k[V]$ such that $f|_{U'} = \frac{g}{h}|_{U'}$ with $h \neq 0$ on U'.
(2) f is regular on U if and only if it is regular at $p \, \forall p \in U$.
(3) Let $\mathcal{O}_V(U)$ be the ring of regular functions on U with pointwise operations.

Theorem 2.39.

$$\mathcal{O}_V(V) \cong k[V]$$

Example 15. $f : U = \mathbb{A}^2 \setminus \mathbb{V}(x) \to k$ *given by* $(x, y) \to \frac{y}{x}$ *is a regular function.*

Definition 2.40. *If $U \subseteq V$ is a quasiprojective variety with a function $f : U \to k$ then f is regular at a point $p \in U$ if $p \in V' \subseteq V$ such that $f|_{V'}$ is regular on V' and V' affine.*

Definition 2.41. *We define $\mathcal{O}_V(U)$ to be the ring of regular function on U.*

Lemma 2.42. *$\mathcal{O}_V(U)$ is a sheaf.*

Definition 2.43. *A function $\psi : V \to W$ of quasiprojective varieties is a morphism of quasiprojective varieties if and only if $\forall p \in V$ if there exist affine opens*

$$\begin{tikzcd} V \arrow[r, "\psi"] & W \\ U \arrow[u, "\iota"] \arrow[r] & U' \arrow[u, "\iota"'] \end{tikzcd}$$

where the bottom arrow is given by regular functions.

Definition 2.44. *An abstract algebraic variety is a topological space X with a sheaf of rings $\mathcal{O}_X$ such that there is an open conver $X = \bigcup U_i$ such that $(U_i, \mathcal{O}_X(U_i))$ are isomorphic (as ringed spaces) to $(W_i, \mathcal{O}_{W_i}(W_i))$ for W_i an affine variety.*

Remark 2.45. *Informally, this is the data of*

$$\{V_\alpha\}_{\alpha \in \Lambda}, \{V_{\alpha,\beta}\}_{\alpha,\beta \in \Lambda}, \{g_{\alpha\beta}\}$$

with V_α affine varieties, $V_{\alpha\beta} \subset V_\alpha$ being open affine subsets and $g_{\alpha\beta} : V_{\alpha\beta} \to\tilde{} \, V_{\beta\alpha}$ such that:

(1) $g_{\alpha\alpha} = \mathrm{Id}_{V_{\alpha\alpha}}$
(2) cocycle condition

Then, $X = \bigcup_\alpha X_\alpha / \sim$ with $x \in X_\alpha \sim u \in X_\beta$ iff $g_{\alpha\beta}(x) = y$.

3. Toric Varieties:

Definition 3.1. *A toric variety is a variety X together with a dense torus $T \subseteq X$ and an algebraic action satisfying:*

$$\begin{tikzcd} T \times X \arrow[r] & X \\ T \times T \arrow[u, hook] \arrow[r] & T \arrow[u, hook] \end{tikzcd}$$

From now on, we assume $k = \overline{k}$.

Definition 3.2. *A group variety is a variety* G *together with morphisms,*

$$\begin{aligned} m : G \times G &\to G \\ i : G &\to G \\ e : * &\to G \end{aligned}$$

i.e. multiplication, inverse, and identity satisfiying the group axioms.
Thus, G *is a group.*

Example 3.3. $k^\times = \mathbb{A}^1 \setminus \{0\}$ *is a group variety. Our maps are,*

$$\begin{aligned} m : (x, y) &\mapsto xy \\ i : x &\mapsto \frac{1}{x} \\ e : * &\mapsto 1 \end{aligned}$$

Proposition 3.4. *Let* G *be an affine variety. Then* G *is a group variety if and only if* $k[G]$ *is a finitely generate commutative* k*-Hopf algebra. That is, it is a finitely generated polynomial ring mod an ideal satisfying the Hopf condition, i.e. there exist maps*

$$\begin{aligned} m^\# : k[G] &\to k[G \times G] \cong k[G] \otimes k[G] \\ i^\# : k[G] &\to k[G] \\ e^\# : k[G] &\to k \end{aligned}$$

satisfying the equivalent group conditions.

Another way to write this is,

Definition 3.5. R *is Hopf algebra if*

$$\begin{aligned} \Delta : R &\to R \otimes R \\ S : R &\to R \\ \varepsilon : R &\to R \end{aligned}$$

called the co-multiplication, antipode, and counit satisfying the group axioms.

In particular, Δ has a nice description,

$$\begin{aligned} \Delta : k[G] &\to k[G \times G] \cong k[G] \otimes k[G] \\ f &\mapsto \Delta(f) = m^\#(f) \end{aligned}$$

is given by

$$\begin{tikzcd} G \times G \arrow[r, "m^\#(f)"] \arrow[d] & k \\ G \arrow[ur, "f"'] & \end{tikzcd}$$

In particular,

$$m^\#(f) : G \times G \to k \qquad m^\#(f)(g, h) = f(gh)$$

Definition 3.6. *An algebraic torus is a variety* T *isomorphic to* $\mathbb{G}_m^n$ *for some* n *where* $\mathbb{G}_m \cong k^\times$.

Remark 3.7. *(1) This is not true if* $k \neq \overline{k}$.
(2) If $T \cong \mathbb{G}_m^n \cong \mathbb{A}^n \setminus \mathbb{V}(x_1, \cdots, x_n)$.
(3) $k[T] \cong k[\mathbb{G}_m^n] \cong k[\mathbb{A}^n \setminus \mathbb{V}(x_1, \cdots, x_n)]$

Moreover, recall,

$$\begin{aligned} k[\mathbb{A}^n \setminus \mathbb{V}(x_1, \ldots, x_n)] &\cong k[x_1, \ldots, x_n]\left[\frac{1}{x_1 \cdots x_n}\right] \\ &\cong k\left[x_1, \ldots, x_n, \frac{1}{x_1}, \cdots, \frac{1}{x_n}\right] \\ &\cong k[x_1^{\pm}, \ldots, x_n^{\pm}] \end{aligned}$$

where this is called the ring of Laurent polynomials.

Lemma 3.8.

$$(k[\mathbb{G}_m^n])^\times = \{c x_1^{m_1} \cdots x_n^{m_n} : c \in k^\times, m_i \in \mathbb{Z}\}$$

Definition 3.9. *A character of a group variety* G *is a group variety map* $\chi : G \to \mathbb{G}_m = k^\times$. *Note that as* $k^\times \cong \mathrm{GL}_1(k)$, *this is a one dimensional representaion.*

Lemma 3.10. *Every character* $\chi : \mathbb{G}_m^n \to \mathbb{G}_m^n$ *is of the form*

$$\chi(t_1, \cdots, t_n) \to t_1^{m_1} \cdots t_n^{m_n}$$

for some $(m_1, \cdots, m_n) \in \mathbb{Z}^n$ *and all such maps appear,*

$$\mathrm{Hom}_{\mathsf{Grp.Var}}(\mathbb{G}_m^n, \mathbb{G}_m) \equiv \mathbb{Z}^n$$

Proof. Let $\chi : \mathbb{G}_m^n \to \mathbb{G}_m$ be a character. This gives,

$$\chi^\# : k[\mathbb{G}_m] \to k[G_m^n]$$

But this is,

$$\chi^\# : k[t, t^{-1}] \to k[x_1^{\pm}, \ldots, x_n^{\pm}]$$

Thus, $\chi^\#$ is determined by where it takes t. Since t is a unit, it must go to a unit, i.e.

$$\chi^\#(t) = c x_1^{m_1} \cdots x_n^{m_n} =: f$$

with $m_i \in \mathbb{Z}$. Thus, we only need to show $c = 1$.

We use the Hopf algebra structure.

$$\begin{aligned} \Delta(f)(g, h) &= f(gh) \\ &= \chi(gh) \\ &= \chi(g)\chi(h) \end{aligned}$$

Thus, $f(gh) = f(g)f(h)$ for all $g, h \in G$. Thus, f is a group homomorphism and $\Delta(f) = f \otimes f$. Thus,

$$\begin{aligned} \Delta(f) &= \Delta(c x_1^{m_1} \cdots x_n^{m_n}) \\ &= c\Delta(x_1^{m_1} \cdots x_n^{m_n}) \end{aligned}$$

However, on the other hand, Δ is a k-algebra map, thus,

$$\Delta(f) = f \otimes f = cx_1^{m_1} \cdots x_n^{m_n} \otimes cx_1^{m_1} \cdots x_n^{m_n} = c^2 \Delta(x_1^{m_1} \cdots x_n^{m_n})$$

Thus, $c^2 = c$ thus $c = 1$.

To complete the proof, show all of these are realized. Not so hard. $\square$

Corollary 3.10.1.

$$\mathrm{Hom}_{GV}(\mathbb{G}_m^r, \mathbb{G}_m^r) \cong \mathrm{Mat}_{r \times n}(\mathbb{Z})$$

That is, if $\phi : \mathbb{G}_m^n \to \mathbb{G}_m^r$ is a group variety map,

$$\phi(t_1, \cdots, t_n) = \left(\prod_{i=1}^n t_i^{a_{1i}}, \ldots, \prod_{i=1}^n t_i^{a_{ni}} \right)$$

That is, maps of tori are monomial maps.

Proof. Composing with the ith projection gives a character,

$$\mathbb{G}_m^n \to \mathbb{G}_m^r \to \mathbb{G}_m$$

$\square$

Corollary 3.10.2.

$$\mathrm{Aut}(\mathbb{G}_m^n) \cong \mathrm{GL}_n(\mathbb{Z})$$

Definition 3.11. *Let T be an algebraic torus. The character lattice of T is*

$$X^*(T) = \mathrm{Hom}_{GV}(T, \mathbb{G}_m)$$

We showed this is a lattice of rank $n = \dim(T)$, where a lattice if a finitely generated free abelian group.

Abstractly, $X^\times(T) \cong \mathbb{Z}^n$ but we write $X^*(T)$ multiplicatively but $\mathbb{Z}^n$ additively.

Definition 3.12. *We introduce the following notational short-hand:*

$$M = X^*(T)$$

This is also the regular functions on the torus.

For $\mathbb{G}_m^n$, we have

(1) $k[\mathbb{G}_m^n] \cong k[x_1^\pm, \ldots, x_n^\pm]$.
(2) $\mathrm{Hom}_{GV}(\mathbb{G}_m^n, \mathbb{G}_m) \cong \{ \text{ Laurent mons. } \} \cong \mathbb{Z}^n$

Thus, $k[\mathbb{G}_m^n] \cong k[\mathbb{Z}^n]$. Moreover,

$$k[\mathbb{Z}^n] \cong \bigoplus_{m \in \mathbb{Z}^n} k \cdot \chi^m$$

where $\chi^m \cdot \chi^n = \chi^{m+n}$.

Thus we are able to identify regular functions on the torus $\mathbb{G}_m^n$ with sums of characters with a direct sum decomposition.

Definition 3.13. *If A is a finitely generated abelian group its group algebra*

$$k[A] \cong \bigoplus_{a \in A} k\chi^a$$

with $\chi^\alpha \cdot \chi^\beta = \chi^{\alpha+\beta}$.

Theorem 3.14. *Let T be an n-dimensional torus. Fix $m_1, \cdots, m_n$ a basis for M.*

(1) Viewing $m_1, \cdots, m_n$ as regular functions on T there is a morphism,

$$T \to^{(m_1,\ldots,m_n)} \mathbb{G}_m^n$$

is an isomorphism of group varieties.

(2) There is an isomorphism of commutative Hopf k-algebras,

$$k[M] \to^{\sim} k[T]$$
$$\chi^m \mapsto m$$

Motto: A torus is recovered from its M lattice.

Where are we heading with this? Say we have X affine with $T \subset X$ a torus. Then, we get a map,

$$k[X] \to k[T]$$

Corollary 3.14.1.

$$(k[T])^\times \cong \{c\chi^m : c \in k^\times, m \in M\}$$

That is the set of invertible regular functions on T are the characters.

Proof. Let $e_1, \ldots, e_n$ be the standard basis for $\mathbb{Z}^n = X^*(\mathbb{G}_m^n)$.

$$e_i(t_1, \ldots, e_n) = t_i$$

Since T is a torus, we may fix an isomorphism

$$\alpha : T \to \mathbb{G}_m^n$$

This induces an isomorphism of rings,

$$\alpha^\# : k[\mathbb{G}_m^n] \to k[T]$$

which is an isomorphism of character lattices contained in these.

Define $u_i = (\alpha^\#)^{-1}(m_i) \in X^*(\mathbb{G}_m^n)$. Since $\alpha^\#$ is and isomorphism and m_i are a basis, we have $u_1, \cdots, u_n$ is a basis for $X^*(\mathbb{G}_m^n) \cong \mathbb{Z}^n$.

Now we have two bases given by the e_i and the u_i for $\mathbb{Z}^n$. Thus ther exists $A \in \mathrm{GL}_n(\mathbb{Z})$ taking $e_i \mapsto u_i$.

Then A is a (monomial) automorphism of $\mu_A : \mathbb{G}_m^n \to^{\sim} \mathbb{G}_m^n$ with the property that $\mu_A^\#(e_i) = u_i$.

Now consider,

$$\begin{tikzcd} T \arrow[r, "\sim", "\alpha"'] \arrow[dr, "\Psi"'] & \mathbb{G}_m^n \arrow[d, "\mu_A"] \\ & \mathbb{G}_m^n \end{tikzcd}$$

We want to show that this diagram commutes. We check this on coordinate rings. Dualizing this picture, we get,

$$\begin{tikzcd} k[T] & k[\mathbb{G}_m^n] \arrow[l, "\alpha^\#", "\sim"'] \\ & k[\mathbb{G}_m^n] \arrow[ul, "\Psi^\#"] \arrow[u, "\mu_A"'] \end{tikzcd}$$

It is enough to check this on $e_1, \cdots, e_n$. We have,

$$\begin{aligned} (\mu_A \circ \alpha)^\#(e_i) &= \alpha^\#(\mu_A^\#(e_i)) \\ &= \alpha^\#(u_i) \\ &= m_i \\ &= \Psi^\#(e_i) \end{aligned}$$

For the second part, we know,

$$\begin{aligned} k[x_1^\pm, \cdots, x_n^\pm] &\to k[M] \\ x_i &\mapsto \chi^{m_i} \end{aligned}$$

Argument similar to before show this is an isomorphism and $k[x_1^\pm, \cdots, x_n^\pm] \cong k[T]$. $\square$